

Application Based Questions - Answers

1. How does cloud federation improve resource sharing and availability across datacenters?

Ans: Cloud federation connects multiple cloud datacenters so that resources can be shared between them. If one datacenter is overloaded or fails, workloads can be moved to another datacenter. This improves resource utilization, ensures high availability, supports load balancing, reduces downtime and provides disaster recovery. Thus, users get reliable and uninterrupted services.

2. What are the key security issues in virtualization, and how can they be mitigated?

Ans: Key issues include VM escape attacks, unauthorized access, hypervisor vulnerabilities, insecure VM migration, and data leakage between VMs. These can be mitigated by using secure and updated hypervisors, enabling strong authentication and access control, encrypting data during storage and migration, isolating VMs using VLANs and firewalls, and regularly applying security patches and monitoring logs.

3. How does a software-defined datacenter enhance scalability and management?

Ans: A software-defined datacenter (SDDC) virtualizes compute, storage and networking resources and manages them through software.

It allows quick resource provisioning, automatic scaling based on demand, centralized management, better resource utilization, reduced hardware dependency and lower operational cost. Hence, it makes the infrastructure more scalable, flexible and easy to manage.

4. How can identity and access control ensure data privacy in cloud-based healthcare systems?

Ans: IAM ensures that only authorized users like doctors, nurses and staff can access patient data. Data privacy can be ensured by using multi-factor authentication (MFA), role-based access control (RBAC), encrypting data, maintaining audit logs, implementing single sign-on (SSO) and following healthcare privacy regulations. This protects sensitive patient information from unauthorized access.

5. How do virtualization and hypervisors support scalability and resource efficiency in cloud applications?

Ans: Hypervisors create multiple virtual machines on a single physical server. This allows better utilization of hardware resources. VMs can be quickly created, moved or removed based on demand. Features like live migration, load balancing and dynamic resource allocation help in scaling applications easily and reduce hardware cost. Thus, virtualization and hypervisors improve scalability, performance and resource efficiency.

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and **COURSE CODE:** CSA15
Big Data Analytics

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Criteria	Relevance to Application (1.25)	Use of Cloud Concepts (1.25)	Practical Insight (1)	Completeness (0.75)	Clarity (0.75)	TOTAL (5)
Weightage	25%	25%	20%	15%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded: